

# Annan Cai

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Dallas, TX

LinkedIn

## EDUCATION

**The University of Texas at Dallas** – M.S., Computer Science

Core GPA: 3.5

*Expected: May 2027*

*Dallas, TX*

Relevant Courses: Machine Learning, Natural Language Processing, Database Design, Big Data Management & Analytics, Systems Security & Binary Code Analysis, Human Computer Interactions, Web Programming Languages

**Civil Aviation University of China** – B.S., Computer Science

*Graduated: June 2025*

*Tianjin, CN*

## SKILLS

**Languages:** JavaScript, Bash, SQL, Python, Java, C/C++, Lua

**Databases:** MongoDB, Redis, MySQL

**Frameworks:** Node, Express, React, Puppeteer, LangChain,

**Tools:** Git, Neovim, Docker, Microsoft Office Suite,

PyTorch, TensorFlow, Gemini API, SocketIO, WebRTC

Google Suite, Cursor, Antigravity, GitHub Copilot

**Soft Skills:** Team Collaboration, Verbal & Written Communication, Analytical Problem Solving, Adaptability

## HACKATHONS & PROJECTS

**HackTX 2025** – Winner, Best Human Intelligence (Out of 728 participants / 225 teams)

*October 18, 2025*

*Austin, TX*

**404 Apparel** – Contributor

- Trained counter-surveillance noise in **PyTorch** to evade YOLOv8 detection, reducing model confidence by **93%**.
- Built a live-testing interface utilizing **OpenCV** and **WebRTC** to validate detection-disrupting capabilities.

**HackUTA 7: Enter the Hacker-Verse** – Winner, Best Use of Agentuity (Out of 429 participants / 135 teams)

*October 4, 2025*

*Arlington, TX*

**Bugzooka** – Contributor

- Engineered an autonomous **QA testing agent** using **Gemini** and prompt engineering, deployed with **Agentuity**.
- Built a backend using **FastAPI** and **MongoDB** via **Beanie ODM**, ensuring type-safe data schemas with **Pydantic**.

**Privacy-Preserving Gateway for LLMs**

*July, 2026*

*Dallas, TX*

- Engineered a privacy-filtering middleware using **Transformer-based NER** to mask PII within user input prompts.
- Optimized redaction accuracy via **Gemini API**, prioritizing high **Recall** to eliminate sensitive data leakage.

**Neural Language Model**

*April, 2026*

*Dallas, TX*

- Architected a language model from scratch in **PyTorch**, manually implementing embedding and softmax layers.
- Trained word embeddings on the **NLTK Brown corpus** using **SGD**, achieving high semantic accuracy on the **WordSim353** benchmark.

## WORK EXPERIENCE

**Shanghai Education Software Development Co., Ltd.** – Software Development Intern

*July 2024 – September 2024*

*Shanghai, CN*

- Participated in front-end development of UI rework, utilizing **Vue** with **Vuetify** component framework.

**Shanghai Supwisdom Information Technology Co., Ltd.** – Software Development Intern

*December 2023 – January 2024*

*Shanghai, CN*

- Migrated frontend codebase from **Vue Options API** to **Composition API**, refactoring **2,000+** lines of code.